Furey Fermions in GALG and FFGFT: N-Operator, Tripotents, and Neutrino-Vacuum Identity

Abstract

Doug Matzke's IPI mail of 3 September 2026 presents concrete GALG results on Furey fermions in Cl(6): complete Up/Down fermion sets Su/Sd, idempotents N_1, N_2, N_3, tripotent N, jE7 projector, and a grade-1+grade-5 construction. The strongest single result: Su$[0]$ = Vss$[0]$ exactly — the neutrino is the lowest vacuum state **[B]**. Theorem F shows: the tripotent property for even vacua and the Frobenius fixed-point property (Doc. 339) are the same decomposition from two independent directions **[B]**. The electron exception at the jE7 projector is the known Furey problem **[S]**.

.1 Theorem A — N_k are idempotents in characteristic 3

$N_k = -1 + (-i\,B_k)$, $B_k^2 = -1$ (Witt pair):

$$N_k^2 = 1 + 2i\,B_k + (-1)(-1) = 2 + 2i\,B_k \equiv -1 + (-i\,B_k) \pmod 3 = N_k. \quad \textbf{[B]}$$

The sum loses its scalar part since $-3 \equiv 0$: $N_1 + N_2 + N_3 \equiv (-i) \sum B_k = N$.

.2 Theorem B — N is a tripotent

Cross-terms $N_k N_j$ ($k \neq j$) are pure grade-4 elements; hence $N^3 = N$ **[B]**. Doug's output: `N**3 = N`, `N**4 = N**2`.

.3 Theorem C — Su$[0]$ = Vss$[0]$: neutrino is the vacuum

`Su[0] == Vss[0]: True` and `(1+N)*(1+jE7) == Vss[0]: True` **[B]**.

FFGFT correspondence [B]. In FFGFT, ν_1 lies in orbit $O_1 = \{1, 3, 9\}$, the orbit of the generator of GF$(27)^*$ — the algebraically simplest object. Both frameworks independently place the neutrino at the algebraically simplest position: convergence of two independent frameworks.

.4 Theorems D–F

D — sigma [B]. $e_1 \to e_2 \to e_4 \to e_1$, $e_3 \to e_6 \to e_5 \to e_3$; `{1:2,2:4,4:1,3:6,6:5,5:3}`, two 3-cycles, $\sigma^3 = $ id. Doc. 341 contains this formula; Doug's `all_g1_g5good` is the σ-invariant sum.

E — jE7 $\equiv N_k$ as projectors [B]; electron exception [S]. $F \cdot jE7 = +F$ for all Su (8/8); $F \cdot jE7 = -F$ for 7/8 of Sd (Sd[7]=electron fails). Same for N_k. Known Furey problem. FFGFT: electron in Dirac layer $\{f_3, f_4\}$, neutrino in Majorana layer $\{f_1, f_2\}$ (Doc. 343, Theorem D‴). Common root open **[S]**.

F — Tripotent for even vacua [B]. $(V_k + P)^2 = -V_k$, $(V_k + P)^3 = V_k$ for $k \in \{0, 2, 4\}$; no collapse for $k \in \{1, 3, 5\}$ (length 32). Identical to the Frobenius fixed-point/orbit split of Doc. 339 — two independent derivations of the same decomposition.

.5 Theorem G — Anti-neutrinos: Dirac or Majorana?

The tension. In GALG/Furey (Theorem C) the neutrino is Su$[0]$ = Vss$[0]$ and the anti-neutrino is Sd$[0]$ = cc(Su$[0]$) = Vss$[1] \neq$ Vss$[0]$. Hence neutrinos in Furey's construction are **Dirac particles** ($\nu \neq \bar{\nu}$).

In FFGFT (Doc. 343, Theorem D‴), ν_1 and ν_2 lie in orbits O_1 and O_3, which are self-inverse under $k \mapsto k^{-1}$ — the *algebraic* Majorana criterion.

Theorem .5.1 (Clarification of Majorana concept **[B]**/**[S]**). *Two distinct Majorana concepts:*

1. ***Algebraic [B]:*** *Orbit O_k is self-inverse under $k \mapsto k^{-1}$ mod 13 — a statement about GF$(27)^*$ structure, not about spin.*
2. ***Physical [S]:*** *$\nu = \bar{\nu}$ in the Lorentz sense (Majorana condition). This requires Sd[0]=Su[0], which does not hold in GALG/Furey.*

*The Galois algebra fixes the algebraic property **[B]**; the physical Majorana condition is an additional assumption **[S]**.*

Position of anti-neutrinos [B]/[S]. GALG: $\bar{\nu}$ = Sd[0] = Vss[1] in the odd vacuum (sigma-orbit, other sector, Doc. 339). FFGFT: the anti-neutrino would lie in the negation $f_1 \leftrightarrow g_1$ (Doc. 343, Theorem D″) — order-13 class g_1, other sector. Both frameworks place the anti-neutrino in the structurally "other" sector.

Consequence for m_{ee}. The prediction $m_{ee} \approx 7.1$ meV from Doc. 340 assumes physical Majorana neutrinos $\nu_{1,2}$. If ν_1, ν_2 are instead Dirac particles (as in Furey), m_{ee} vanishes entirely. The m_{ee} prediction is conditioned on the Majorana assumption **[S]**.

Open questions [S]. (1) Common algebraic root of the electron inconsistency in GALG/Furey and the Majorana/Dirac layer in FFGFT. (2) Whether ν_1, ν_2 are physical Majorana particles is not decided by the Galois algebra — this is the central open bridge between GALG/Furey and FFGFT.

.6 Summary

Table 1: Results Doc. 344

Theorem	Content	Status
A	$N_k^2 = N_k$, characteristic-3 proof	**[B]**
B	$N^3 = N$, tripotent	**[B]**
C	Su[0]=Vss[0]: neutrino=vacuum; FFGFT correspondence	**[B]**
D	sigma= $\{1 \to 2, 2 \to 4, 4 \to 1, 3 \to 6, 6 \to 5, 5 \to 3\}$	**[B]**
E	jE7$\equiv N_k$; electron exception (Furey problem)	**[B]/[S]**
F	Tripotent even vacua = Frobenius fixed points (Doc. 339)	**[B]**
G	Anti-neutrino Sd[0]=Vss[1]≠Su[0]; algebraic Majorana ≠ physical Majorana	**[B]/[S]**

Open question [S]: Common algebraic root of the electron inconsistency in GALG/Furey and the Majorana/Dirac layer separation in FFGFT (Doc. 343, Theorem D‴).

Verification script

`pruef_344_furey_galg.py` — 6 theorems, all assertions passed.

Bibliography

[1] C. Furey, *Standard Model Physics from an Algebra?* (2016).

[2] D. Matzke, IPI mail 3 September 2026.

[3] J. Pascher, *Doc. 339: Frobenius separation*, FFGFT corpus (Aug. 2026).

[4] J. Pascher, *Doc. 341: GF(27) in GALG and FFGFT*, FFGFT corpus (Sept. 2026).

[5] J. Pascher, *Doc. 343: Zeta function of* T^4/Z_3, FFGFT corpus (Sept. 2026).